TOUCH-ACTIVATED PERSONAL SAFETY

SparshaGuard

Sparśa · Kannada for touch

The touch itself raises the alarm.

A 4G smartwatch that calls her family and 112 when a stranger grabs her. No phone. No button.

- Own 4G SIM
- 15-second PIN
- Duress PIN



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01 · THE PROBLEM

Every SOS tool needs her to act first.

4,48,211

crimes against women recorded in India in 2023

51

every hour, around the clock

40%

of urban women feel unsafe or not so safe

TO CALL FOR HELP TODAY, SHE MUST

1 Find the phone → 2 Unlock it → 3 Open the app → 4 Press SOS

During an attack, her hands are not free, and the phone is out of reach.

● = 1,000 cases ● assault to outrage her modesty: 83,891, about 1 in 5

NCRB, Crime in India 2023 (released Sep 2025): 4,48,211 cases; Sec. 354 IPC 83,891. NARI 2025 (National Commission for Women): 12,770 women, 31 cities.

02 · THE SOLUTION

The watch notices the attack by itself.

- Her own hand: ignored
- Watch ripped off: instant alert

1 **Guard Mode on**
She switches it on when she feels unsafe. Otherwise, it is a normal smartwatch.

2 **A stranger's touch → 15-second PIN**
A grab, a pull or a struggle. It vibrates, starts recording and asks for her PIN.

3 **No PIN → family + 112 alerted**
Her live GPS location and audio go out over its own 4G SIM.

4 **Forced to cancel? Duress PIN**
The screen says "Cancelled" while the alert is sent silently.



03 · HOW IT KNOWS

Two signals tell a stranger from her own hand.

SIGNAL 1 · TOUCH SWEEP

Is a hand on the strap?

A tiny signal sweeps through the strap. Any hand on it moves the reading by **~16%**, against **0.1%** noise.



SIGNAL 2 · STATIC CHARGE

Is it another body?

Every body carries its own static charge. When another person touches her, it evens out and her voltage **jumps**.



	TOUCH SWEEP	STATIC CHARGE	THE WATCH
Her own hand	changes	flat: same body	stays quiet
A stranger	changes	jumps	asks for PIN
A struggle	—	arm jerks (motion)	asks for PIN

↑ Steel case back, on her skin

Strap electrode

Built on **72** published sources

Cross-checked by **2** independent solvers: ngspice, MIT FastCap

04 · THE PRODUCT

A flagship 4G smartwatch, with a safety core inside.

- Own nano-SIM: 4G VoLTE calls and SMS
- Built-in GPS, live location
- 1.43″ AMOLED, always-on, crown
- Heart rate · SpO_2 · ECG · skin temperature
- Wi-Fi 6 · Bluetooth 5.3 · NFC
- 500 mAh · wireless charging

Two processors. One runs the apps; a separate **safety core** runs Guard Mode alone, so a frozen app can never block an alert. Our own SparshaOS, on open-source Zephyr.



- Design specification · hardware prototype next

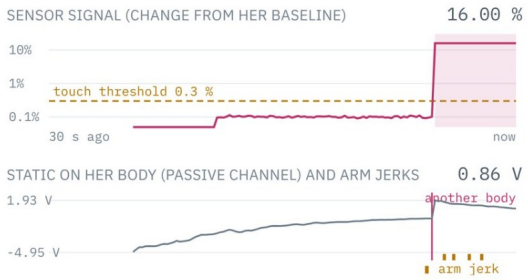
05 · PROOF

Tested on a digital twin before any hardware.

The watch's real software, running on a simulated body. Live on our website.



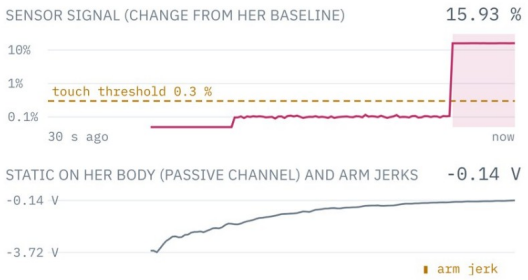
Enter PIN: 11 s left



● A stranger grabs her wrist: both signals jump → **PIN**



Touch on the watch, but no other body: ignored



● Her own hand: static stays flat → **ignored**

90%

of grabs caught across 7 places on her body
(strap sweep alone: 13%)

100%

of struggles caught by the motion sensor

2

unwanted PIN prompts in 9 simulated hours

100%

of strap removals caught

• bhajanth13.github.io/SparshaGuard

Simulation results, not yet measured on hardware. The Gate 1 battery prototype will calibrate them against real body voltages.

06 · MARKET & COMPETITION

A huge market, and no one detects the touch.

9.89 crore
urban women aged 20–44

28.9 M
smartwatches sold in India in 2025

40%
of urban women feel unsafe

PRODUCT	PRICE	WORKS WITHOUT A PHONE	RAISES THE ALARM BY ITSELF
Budget SOS watches Noise, boAt	₹1,999–2,699	✕ needs the phone	✕ press a button
Children's 4G watches Pebble Junior	₹5,999	✓ own SIM	✕ press SOS
Premium LTE watches Apple, Samsung	₹41,999–51,499	✓ own LTE	◖ only a hard fall or crash
SparshaGuard Band · Watch	**₹11,999 · ₹24,999**	✓ own 4G SIM	**✓ another person's touch, a pull or a struggle**

The closest idea is an MIT research sticker, never sold. **No product in India raises an alert on another person's touch.**

MoSPI, Women and Men in India (population projection); IDC India wearables, 2025; NARI 2025. Prices as listed by brands and retailers.

07 · BUSINESS MODEL

Two products, four revenue streams, one free promise.

SPARSHAGUARD BAND

₹11,999

The safety system alone: 4G, GPS, PIN screen, sensing.

Making cost ≈ ₹6,500*

SPARSHAGUARD WATCH

₹24,999

The full flagship smartwatch, with Guard Mode built in.

Making cost ≈ ₹17,000 → ₹11,500 at scale*

GUARD+ FAMILY APP

₹99 /month

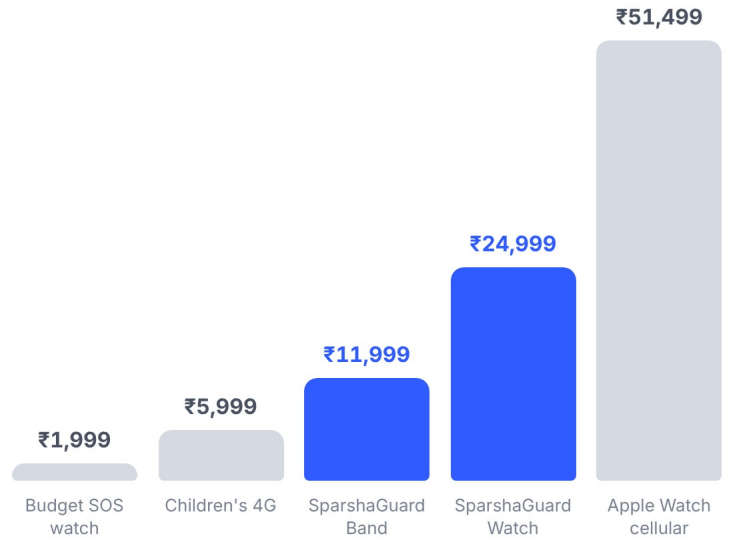
Live map during an alert, encrypted recordings, alert history.

CAMPUS & WORKPLACE

₹4,500 /user/year

Band, Guard+ and replacements, a safety dashboard, 3-year term.

PRICE LADDER · BARS TO SCALE



- Calling for help: free for life
- We never sell location or audio data

* Our estimates from typical distributor prices at 1,000 units (₹11,500 at 10,000 units), to be confirmed with supplier quotes. Prices include GST.

08 · GO-TO-MARKET & ROADMAP

Start where safety is bought in bulk.

PHASE 1 · YEAR 1

Campus pilot

200 users in colleges and hostels in Karnataka, starting at RV University.

Paid by the institution or parents

PHASE 2 · YEAR 2

Employers

IT, BPO, healthcare and retail firms with women on night shifts.

Paid by the employer

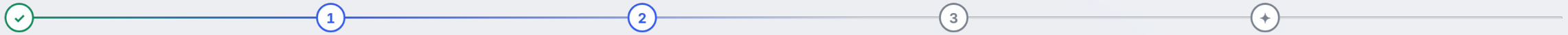
PHASE 3 · YEAR 3

Online and retail

Direct to women and their families, on our site and marketplaces.

Paid by the customer

HARDWARE ROADMAP



DONE
Simulation study + digital twin
72 sources · 2 solvers

GATE 1
Battery bench prototype
Measures real body voltages

GATE 2
Wearable SparshaGuard Band
Custom SMT board, own SIM

GATE 3
Full SparshaGuard Watch
Two processors, AMOLED

LAUNCH
Indian certification
WPC · TEC/MTCTE · BIS

Thank you.

SparshaGuard: the touch itself raises the alarm.

We welcome your questions.



PRESENTED BY

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Try the digital twin
bhajanth13.github.io/SparshaGuard